Predictive and Generative AI

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Predictive and Generative AI

> Prepare Predictive and Generative AI: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary **target**: Distinguish discriminative **prediction** from generation by inspecting the requested **output**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-2/27, ai-3/1. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Capstone Expo

Without opening the old lesson, explain one key idea from Capstone Expo. Give an example and a mistake that the idea helps you avoid.

Look for: Defend an application using reproducible **evidence**

Reference example, not the only acceptable wording: Fictional raw dataset, schema, code version, settings, run instructions, tests with expected/actual results, rejection reasons and change log.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A study tool finds a paragraph, labels its difficulty and invents practice questions. Identify the three operations and one check for each.

Concept to explain

Week 1: Predictive and **Generative AI**

Term	Meaning
predictive AI	AI used to estimate a target such as a category, number or future outcome from inputs.
generative AI	AI that produces new content such as text, images or audio rather than only selecting a category.
target	The category, value or outcome a prediction task is intended to estimate.
output space	The set or range of outputs a system can produce for a task.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

For the compact core route, use two 45-minute sessions. All model **output**s and probability tables are constructed classroom examples. No hosted AI service, external API or student account is needed.

Distinguish a model objective from factual support and system permissions. Collect independent work before discussing its solutions.

Readiness answer

A **prediction** is a model **output**. Truth requires appropriate independent **observation**s or sources; confidence and low **training** loss do not supply that **evidence**.

Printed lesson and workbook; paper and pencil

Calculator; optional local Python 3 interpreter

Prepare the independent task and fresh retry without displaying solutions

End goal for the pictorial case

By the end, I can identify generation and **prediction** objectives without confusing either with verified truth.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Predictive and Generative AI using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Classifier: choose one label from cat, dog, or rabbit.

2. Regressor: estimate a numerical reading time.

3. Generator: construct a new explanation token by token.

4. A product may combine **mechanism**s, so test each stage separately.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

The teaching case

A tool labels a question biology, writes a new question and computes 7 times 8.

The teaching case

Identify each operation.

Exact explanation to model

> **Why the result follows**
>
> Labelling selects a category; generation creates text; calculation returns 56. Generated questions need factual and instructional checks.

Explain the added worked example

Explain the added worked example · 1

A classifier might map a message to the label request or report. A text generator instead produces a sequence, predicting a distribution over the next token given a prefix.

Both make **prediction**s, but the **training target**s and permitted **output**s differ. One application can combine generation with a classifier, retrieval and human approval.

Calling a whole product “generative” does not describe every component.

The objective defines what **training** attempts to fit. Next-token fit rewards matching observed continuations; it does not directly certify that an answer is true, well sourced or permitted.

A fluent completion is still an **output** to evaluate against the task.

Explain the added worked example · 2

There are ten observed continuations. The empirical probabilities are red 6/10 = 0.6, blue 3/10 = 0.3, green 1/10 = 0.1. They are nonnegative and sum to 1.

A simple frequentist classroom model uses these relative counts as its next-token distribution. These probabilities describe the supplied corpus counts, not the current colour of any actual marker.

Key terms

Term	Meaning
Generative objective	A training aim concerned with producing or modelling possible outputs
Discriminative objective	A training aim mapping inputs to a specified target such as a label

Term	Meaning
Conditional distribution	Probabilities for alternatives given stated information
Autoregressive generation	Producing a sequence one next item at a time using preceding items

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Model the reasoning aloud · 3

After producing red, the next step conditions on the expanded prefix “The marker is red”. A second table might give an end token probability 0.8 and the token today probability 0.2.

A generated sequence can stop at the end token or continue. Do not reuse the first-step table unchanged after adding a token unless the model explicitly ignores the new prefix.

Model the reasoning aloud · 4

A separate classifier can return red as a label from an image; that is a different **input** and **target** from continuing text. A downstream evaluation asks whether the system correctly reports the current marker.

It must include ground truth or sources and appropriate failure handling, not just count plausible phrases.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Predictive and Generative AI

A fictional learning tool first classifies a question, then drafts an answer from a supplied note.

The facts we will use

> **Supplied facts**
>
> A fitted classifier labels a question science; its **training** used labelled questions.

> **Supplied facts**
>
> Approved note N: evaporation changes liquid water into water vapour. A generator drafts: “Water evaporates into vapour and always disappears forever.”

AI ACADEMY / AI-3/1

Predictive and Generative AI

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Predict category

The classifier predicts science from the new question.



2. Retrieve note

N supplies a specific statement about evaporation.



3. Generate draft

The draft adds "always disappears forever" beyond N.



4. Verify claims

Keep the supported change-of-state claim; remove or review the unsupported addition.

CHANGE ONE THING

The classifier instead predicts history while N and the draft stay unchanged.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Predict category	The classifier predicts science from the new question.
2. Retrieve note	N supplies a specific statement about evaporation.
3. Generate draft	The draft adds “always disappears forever” beyond N.
4. Verify claims	Keep the supported change-of-state claim; remove or review the unsupported addition.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Classification predicts a category under a trained mapping. Generation produces a sequence of words under a language objective. Supplying a note can support factual grounding, but the draft still requires claim-level checking.

The category being correct does not prove every sentence in the answer is correct.

What this example does not establish

Keep the stated limit

This example does not expose a real model’s architecture or prove that source-fed generation always stays grounded.

Change one thing in the pictured case

Change one thing

Change: The classifier instead predicts history while N and the draft stay unchanged.

Predicted result

Prediction: The category **prediction** changes, but the same unsupported draft claim still needs checking.

Why the result follows

Reason: Stages can fail independently; one result does not validate another.

Ask while showing the picture

Ask while showing the picture · 5

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Predictive and Generative AI, reduce support deliberately and check individual reasoning before discussing answers.

Choose the pacing route

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Board sequence

1. Put the facts on the board

Write the given facts: A tool labels a question biology, writes a new question and computes 7 times 8.

2. State the decision to explain

Write the question to explain: Identify each operation.

3. Reveal the worked reasoning

Reveal after **prediction**s: Labelling selects a category; generation creates text; calculation returns 56. Generated questions need factual and instructional checks.

Hold back the independent task

Keep for an independent attempt: A system labels a message as spam. Another writes a new reply. Which task predicts a category and which generates content? Can the new reply contain a false fact?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: open 0.8 and closed 0.2. The corpus prefers open, but the sensor supplies current closed **evidence**. Exit: a conditional continuation probability is not verification of a current fact.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and distinguish a numerical claim from its interpretation.
7 to 18 minutes	Read the explanation in “Notice the evidence” and “See how it works”, then complete W1; ask for a reason alongside each answer.
18 to 32 minutes	Trace the “A closer explanation” material in “See how it works” and “Follow a worked example”, then complete W2. The pictorial case is an additional example.
32 to 45 minutes	Discuss the diagram and complete W3 in pairs; record support.
Session 2 start to 8 minutes	Complete the guided response before sharing the answer.
8 to 20 minutes	Read the remaining sections and trace another constructed case.
20 to 35 minutes	Collect W4 independently, requiring calculations and a bounded conclusion.
35 to 45 minutes	Discuss W5 and the exit response. Provide feedback and a fresh retry.

Reduce help deliberately

Completion practice

Completion task: For The marker is, calculate the three empirical probabilities, their sum and their meaning.

Reasoning to listen for

Teacher check: Teacher explanation: Probabilities 0.6,0.3,0.1 sum to 1 and describe ten supplied corpus continuations; they are not **observation**s of the current marker.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Generative and predictive objectives. Proposed explanation: “Generative models do not predict”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **Expected repair**
>
> Expected correction: **Autoregressive generation** predicts the next item repeatedly.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short **target**ed group explanation while others solve a more demanding variation.

Pictorial practice question

Identify the first unsupported claim and explain what source would be needed before asserting it.

Reasoned teacher answer

“Always disappears forever” is unsupported and misleading as phrased; do not assert it. The approved note supports liquid changing into vapour, not disappearance.

Changed-case reasoning: The category **prediction** changes, but the same unsupported draft claim still needs checking. Stages can fail independently; one result does not validate another.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Predictive and Generative AI investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Sort twelve systems by **mechanism**. For each, name its **output space**, **evidence** needed, and one possible failure.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Generative models do not predict: **Autoregressive generation** predicts the next item repeatedly.

The most frequent phrase is the current truth: Corpus frequency and current **observation**s answer different questions.

A single accuracy score evaluates the whole system: Generation, retrieval, grounding, formatting and permissions require appropriate separate checks.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Predictive and Generative AI questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Labelling selects a category; generation creates text; calculation returns 56. Generated questions need factual and instructional checks.

Independent question

Independent question · withhold the key

A system labels a message as spam. Another writes a new reply. Which task predicts a category and which generates content? Can the new reply contain a false fact?

Expected independent answer

Teacher key · Expected independent answer

Spam labeling predicts a category; writing the reply generates content. Generated wording can contain false claims and needs checking.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Fluent new wording proves the reply is factual.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

> **Teacher key · Original transfer answer**
>
> Retrieval needs the correct paragraph; **prediction** needs a defined difficulty label and suitable evaluation; generation needs factual and instructional checks.
>
> A correct retrieved passage does not guarantee appropriate questions.

Explanations for the additional workbook examples

W1 Identify the objective

Workbook question · W1

Classify these components by **target**: label a message request/report; continue a prefix with a sequence; retrieve a stored sensor record. Explain why the application may combine them.

Teacher reasoning · W1

Teacher explanation: Message labels are discriminative **prediction**; continuing a sequence is generation; record lookup is retrieval rather than learned generation by itself.

One application can route a request, retrieve **evidence** and generate a response.

W2 Compute the distribution

Workbook question · W2

For The marker is, calculate the three empirical probabilities, their sum and their meaning.

Teacher reasoning · W2

Teacher explanation: Probabilities 0.6,0.3,0.1 sum to 1 and describe ten supplied corpus continuations; they are not **observation**s of the current marker.

W3 Follow the prefix

Workbook question · W3

After red is selected, the expanded-prefix table gives end 0.8 and today 0.2. Explain why these values cannot be substituted for the red/blue/green probabilities.

Teacher reasoning · W3

Teacher explanation: The conditional prefix and alternatives differ. The second table describes what follows The marker is red, not what colour follows the earlier prefix.

W4 Analyse a new corpus independently

Workbook question · W4

For prefix The signal is, counts are high 2, low 5, unknown 3. Compute all probabilities and the most frequent continuation. A current measurement reports high.

State what the most frequent continuation does and does not establish.

Teacher reasoning · W4

Teacher explanation: Probabilities high 0.2, low 0.5, unknown 0.3; low is most frequent. This describes the constructed corpus, not the current signal.

Current high is supported by the supplied measurement; generating low would conflict with that **observation**.

W5 Design a useful evaluation

Workbook question · W5

A system reports current room status using retrieved measurements and generated sentences. Name checks for **evidence**, response quality and permission.

Teacher reasoning · W5

Teacher explanation: Check the correct room and measurement time/source, agreement of every status claim with retrieved data, required response format and handling of missing/conflicting **observation**s, and that only approved read/report actions occur.

Continuation fit alone is insufficient.

Assess independent transfer

Independent question · withhold the key

Selected case: A system retrieves an approved Moon note, uses a trained classifier to predict reading difficulty, then writes a poem. Identify each **mechanism** and the **evidence** needed to check its **output**.

Selected case answer

Teacher key · Selected case answer

Retrieval finds an existing note; classification predicts from fitted patterns; generation creates text.

Check relevance and provenance of the note, difficulty **prediction**s against suitable held-out labels, and poem claims against sources. Fluent style is not factual **evidence**.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

> Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Predictive and Generative AI to Verified Study Buddy, adapt support and plan a later changed-case check.

Verified Study Buddy

Separate Study Buddy tasks into classify, calculate, retrieve, and generate stages.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 2: A Distribution of Possibilities. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide labelled intermediate columns and allow a calculator. Ask students to state the meaning and denominator of a value before interpreting it.

Extension: implement the classroom calculation locally and test boundary cases without using personal data.

W4 requires 0.2,0.5,0.3 and low, with the corpus/current-**evidence** distinction. Fresh retry: counts ready 7, busy 1, unknown 2 and current **observation** busy.

Probabilities 0.7,0.1,0.2; ready is most frequent but does not establish the current state.

Week 2 defines a small token vocabulary and traces encoding and decoding without assuming tokens are words.

Plan the delayed return

Suggested return points in this level: ai-3/2, ai-3/4, ai-3/7. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	

Record	Teacher evidence
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For the Verified Study Buddy, separate routing **prediction**s, source retrieval and generated explanations in its **evidence** record.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Predictive and Generative AI lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching **target**: Distinguish discriminative **prediction** from generation by inspecting the requested **output**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: For The marker is, calculate the three empirical probabilities, their sum and their meaning.

Supported practice answer

Reasoned practice answer: Teacher explanation: Probabilities 0.6,0.3,0.1 sum to 1 and describe ten supplied corpus continuations; they are not **observation**s of the current marker.

Check every learner before discussion: Use the worked case for Generative and predictive objectives. Proposed explanation: “Generative models do not predict”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: **Autoregressive generation** predicts the next item repeatedly.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: The classifier instead predicts history while N and the draft stay unchanged. Require a **prediction** and a reason before comparing.

Expected result: The category **prediction** changes, but the same unsupported draft claim still needs checking. Reason: Stages can fail independently; one result does not validate another.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

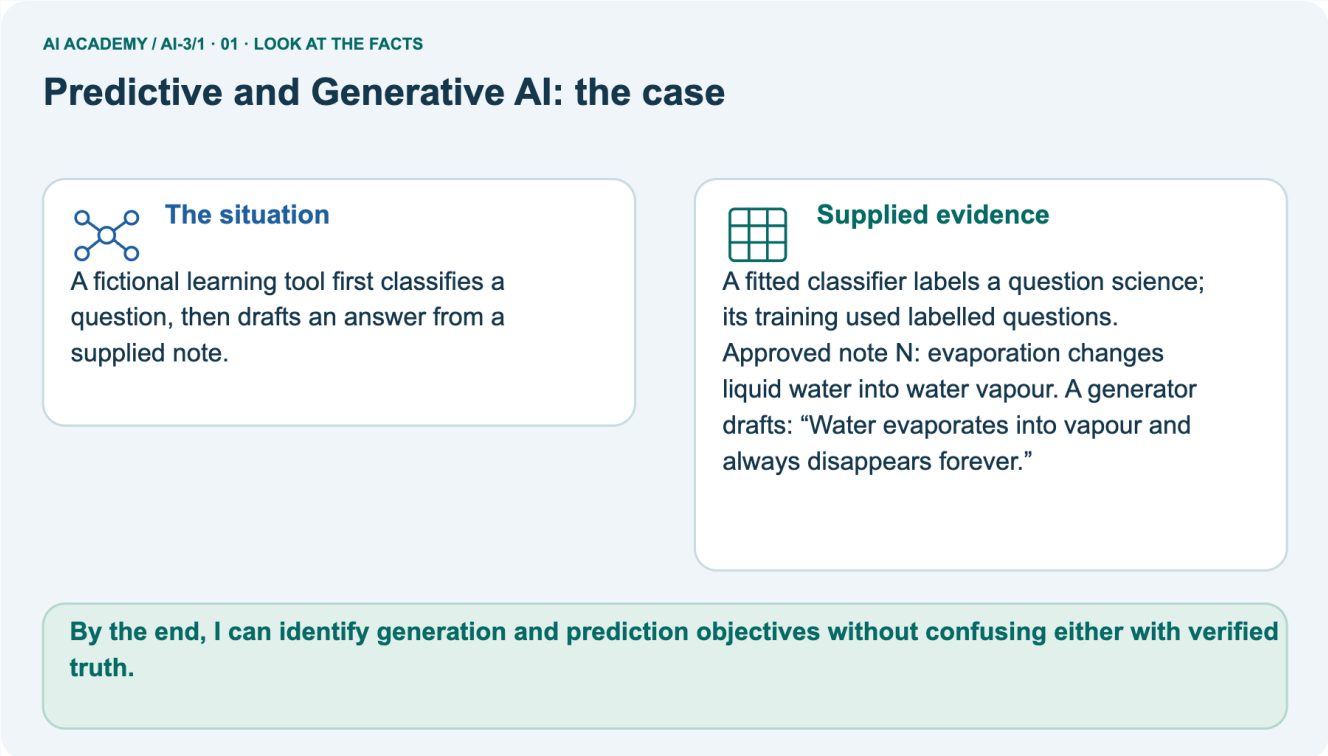
TEACHER GUIDE · 9 OF 9

Visual concept journey

> Show one Predictive and Generative AI graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Predictive and Generative AI: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional learning tool first classifies a question, then drafts an answer from a supplied note. Supplied facts: A fitted classifier labels a question science; its **training** used labelled questions.

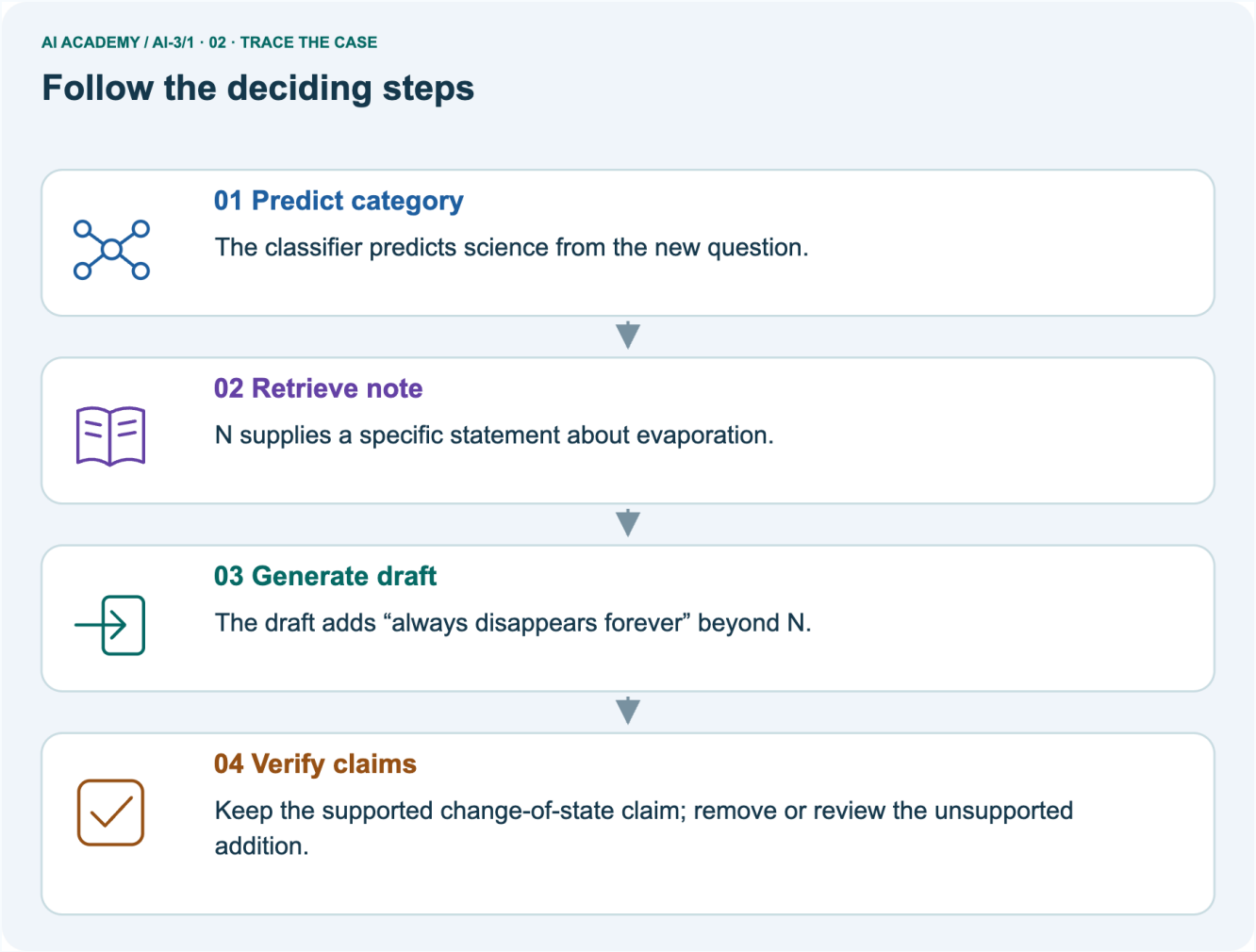
Approved note N: evaporation changes liquid water into water vapour. A generator drafts: “Water evaporates into vapour and always disappears forever.”

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

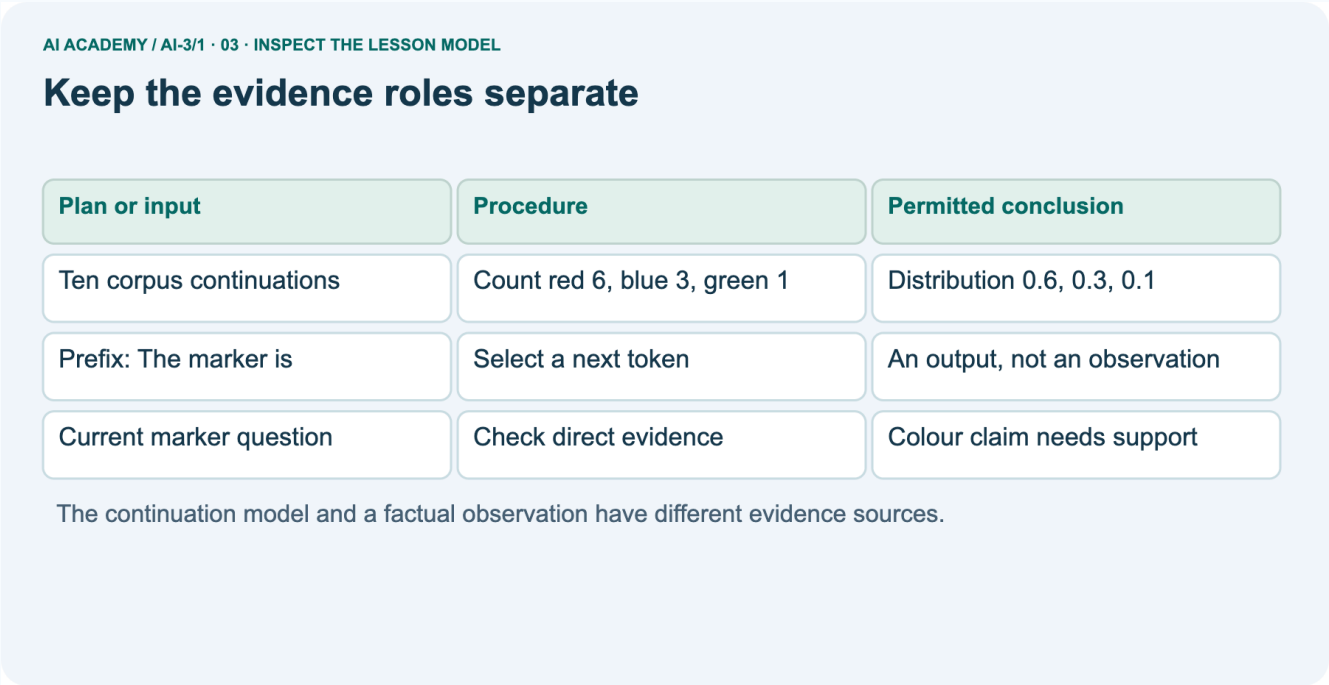
Predict category: The classifier predicts science from the new question. → Retrieve note: N supplies a specific statement about evaporation. → Generate draft: The draft adds “always disappears forever” beyond N. → Verify claims: Keep the supported change-of-state claim; remove or review the unsupported addition.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Keep the evidence roles separate



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the **evidence** roles separate. Ten corpus continuations | Count red 6, blue 3, green 1 | Distribution 0.6, 0.3, 0.1. Prefix: The marker is | Select a next token | An **output**, not an **observation**.

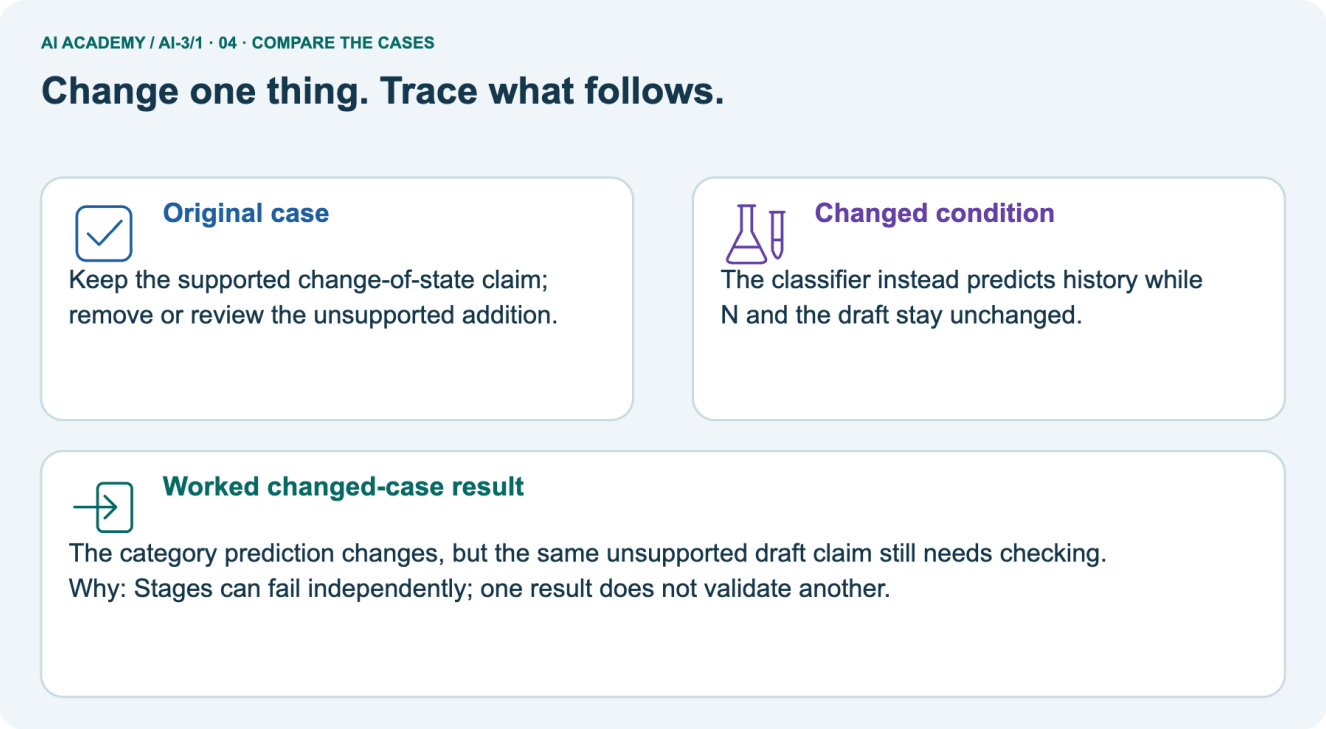
Current marker question | Check direct **evidence** | Colour claim needs support

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Keep the supported change-of-state claim; remove or review the unsupported addition. Changed condition: The classifier instead predicts history while N and the draft stay unchanged.

Expected result: The category **prediction** changes, but the same unsupported draft claim still needs checking. Reason: Stages can fail independently; one result does not validate another.

Predict before checking the worked changed-case result. Name what changed and what the

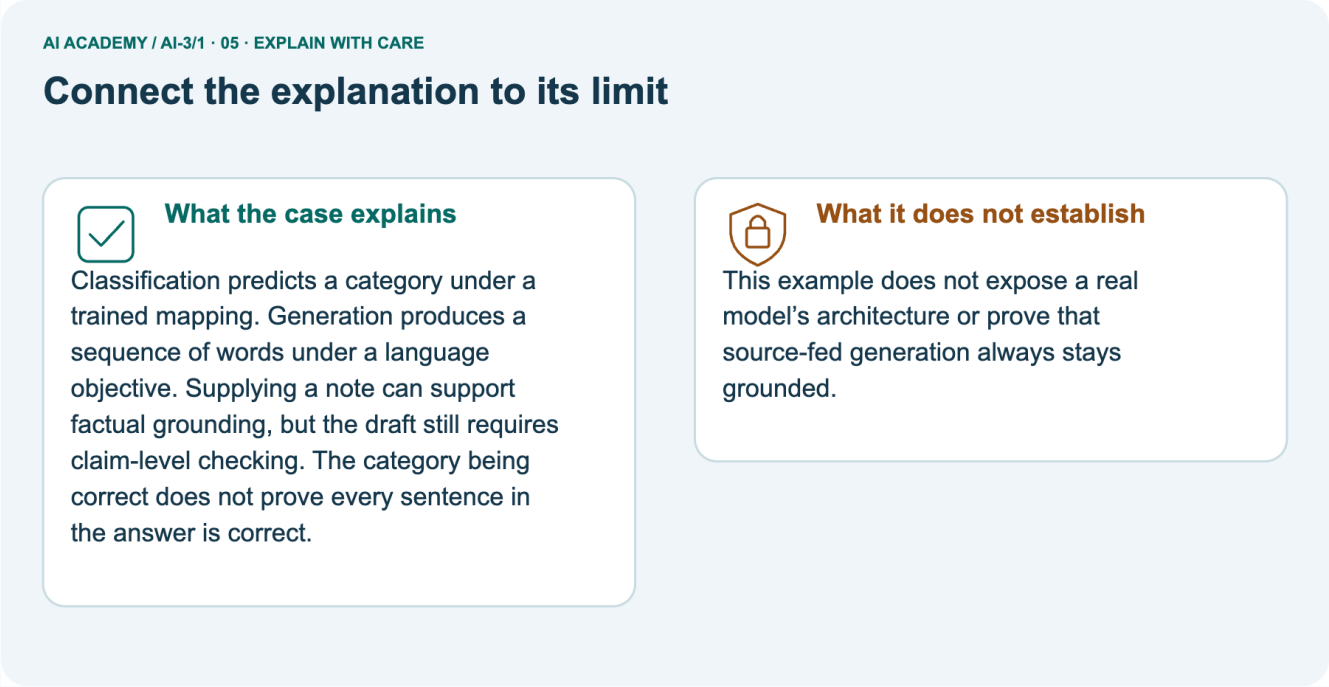
supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Stages can fail independently; one result does not validate another.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Classification predicts a category under a trained mapping. Generation produces a sequence of words under a language objective. Supplying a note can support factual grounding, but the draft still requires claim-level checking.

The category being correct does not prove every sentence in the answer is correct. Limit: This example does not expose a real model’s architecture or prove that source-fed generation always stays grounded.

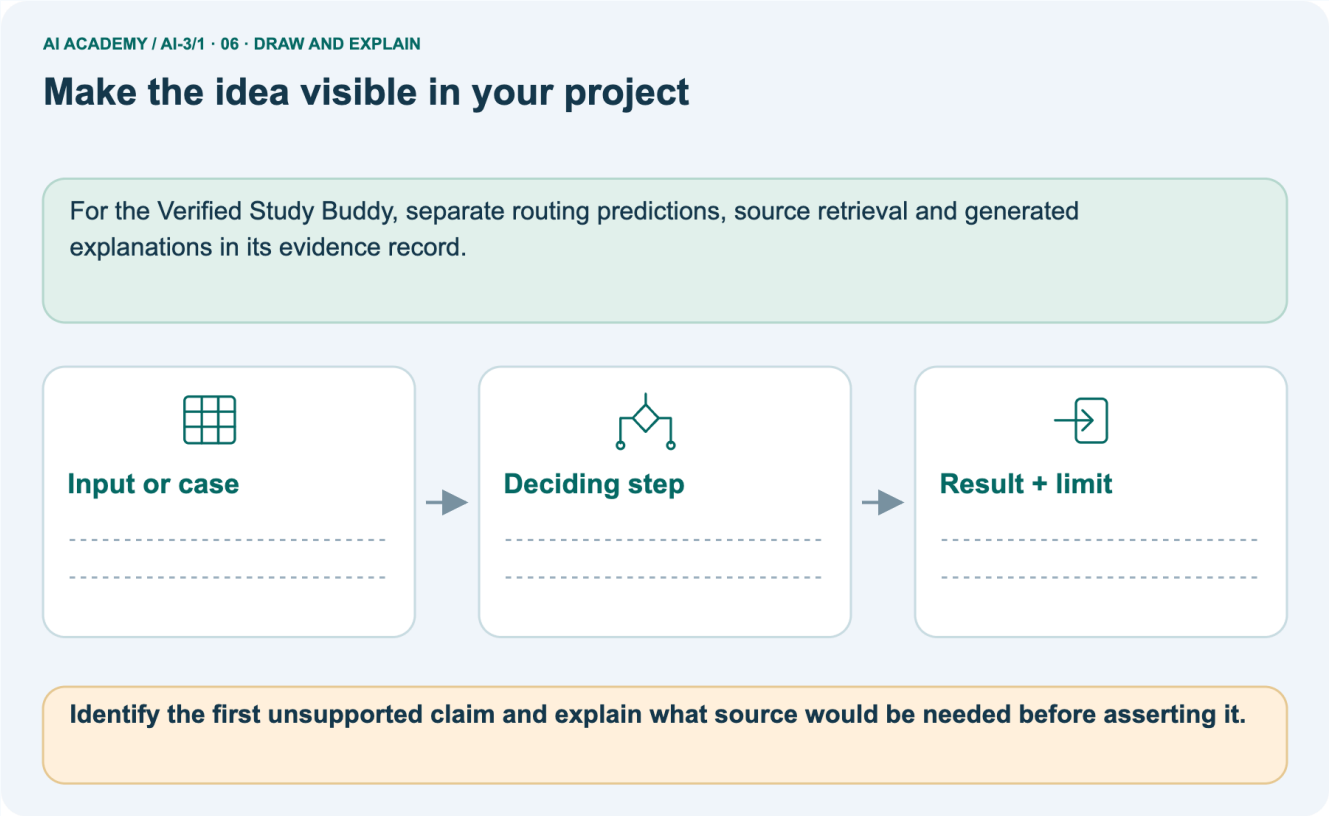
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This example does not expose a real model’s architecture or prove that source-fed generation always stays grounded.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Verified Study Buddy, separate routing **prediction**s, source retrieval and generated explanations in its **evidence** record. Draw your **input**, deciding step, result and limit.

Supported question: Identify the first unsupported claim and explain what source would be needed before asserting it.

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Supported practice key: “Always disappears forever” is unsupported and misleading as phrased; do not assert it. The approved note supports liquid changing into vapour, not disappearance.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.